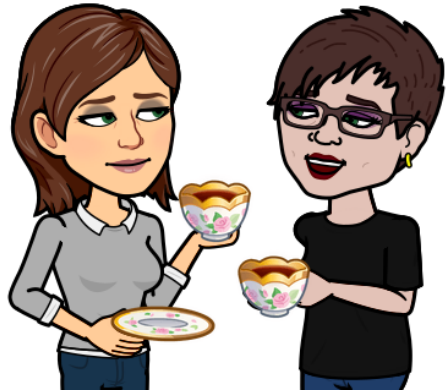


Get to know you!

Please complete the
survey and return to
Megan before you
leave.





MAS!

PPE-

When you come to lab,
what should you put on?

- ▶ Please review the item cards.
- ▶ Can you think of when you will need each of the items?



PLEASE GRAB A HANDOUT TO
REVIEW AND SIGN.



CAN YOU THINK OF A RULE
TO ADD TO THE CLASS LIST?

Room Safety-

MSDS

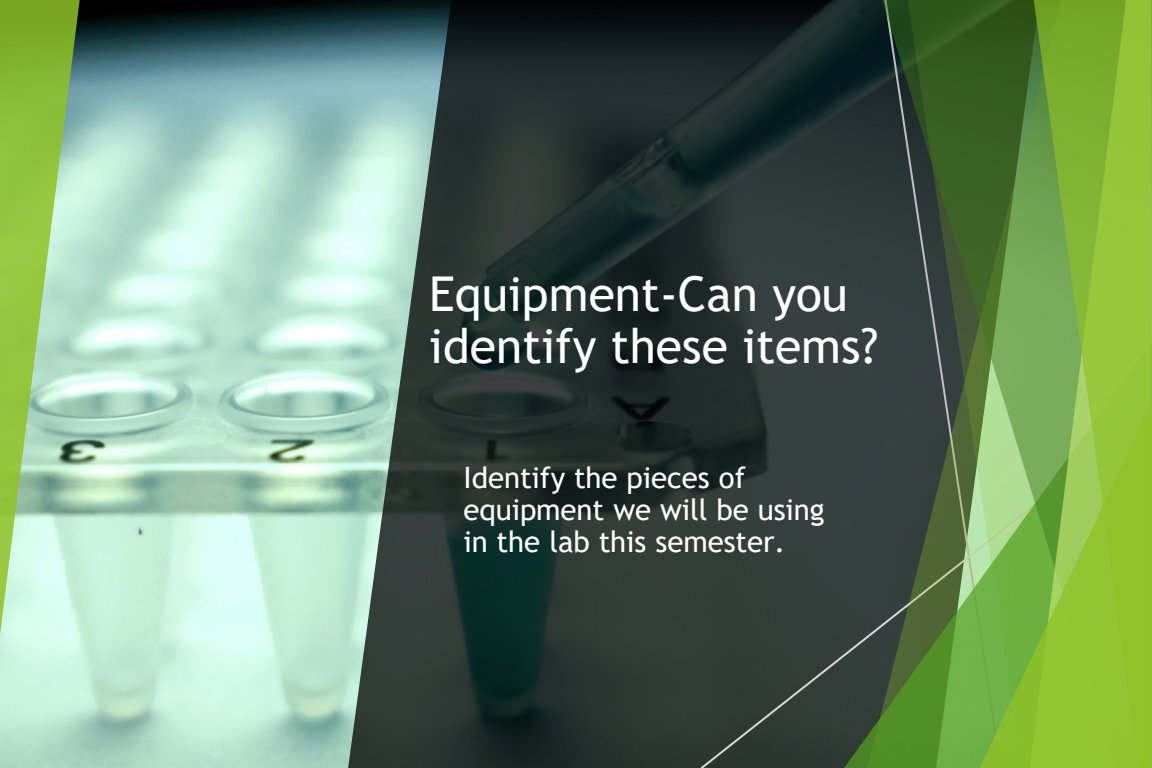
- ▶ Review the MSDS.
- ▶ Complete the pertaining worksheet.



Lab Safety-

Please review the cartoon and complete the corresponding worksheet.





Equipment-Can you identify these items?

Identify the pieces of equipment we will be using in the lab this semester.

Sci-fi movie



Please add your favorite sci-fi movie to the list.



Can you think of one that showcases this course?

Petri film- Swab the school.

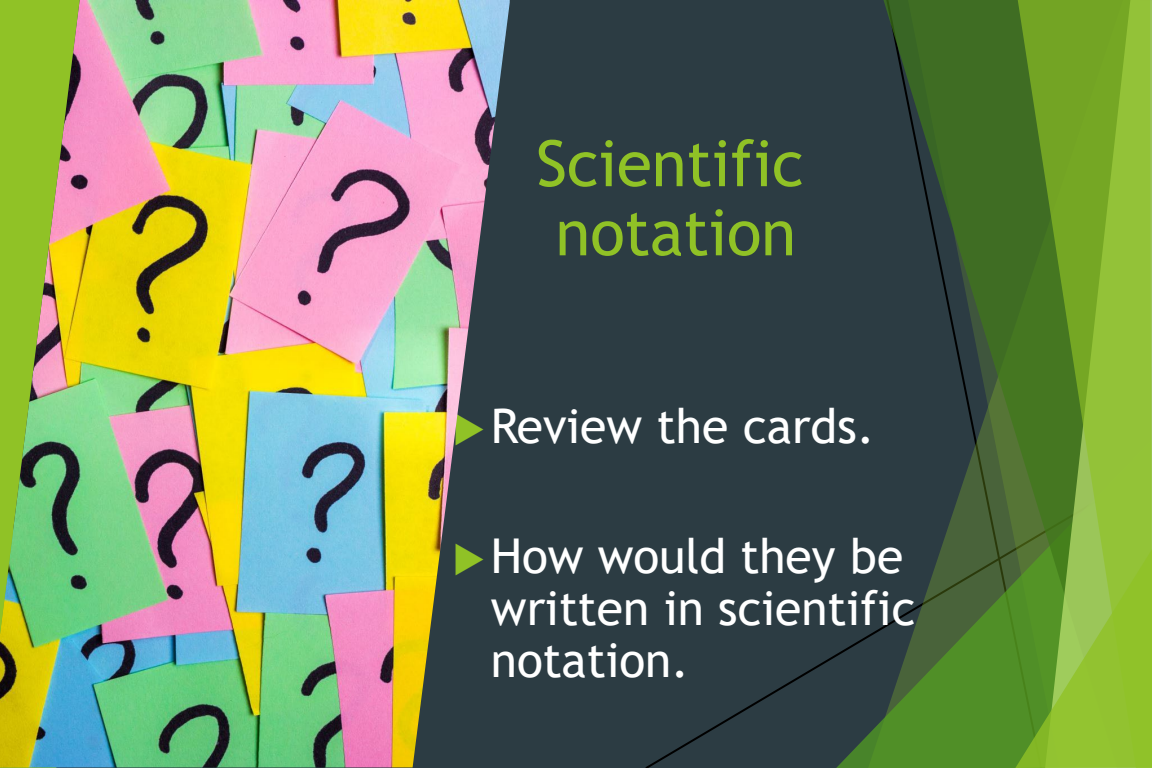
- ▶ Please review the insert for petri film.
- ▶ Swab a few items in the classroom inculcate the period film.





Microscope

- ▶ With your partner describe the steps for viewing the slide on the microscope.
- ▶ What is on the slide?

The background of the slide is a collage of overlapping, torn-edge sticky notes in various colors: pink, yellow, light blue, and light green. Each sticky note has a large, bold black question mark printed on it. The sticky notes are scattered across the left side of the slide, creating a textured, layered effect. The right side of the slide is a solid dark grey-blue area with a subtle geometric pattern of overlapping triangles in shades of green and blue.

Scientific notation

- ▶ Review the cards.
- ▶ How would they be written in scientific notation.



Concentration-Dimensional Analysis

- ▶ Review the cards.
- ▶ Have you seen this before?
- ▶ How would you solve?
- ▶ Does it scare you?

Table of the Element

Periodic table

H ← Symbol Hydrogen 1.008 ← Atomic Weight																	
7 IIB			8 VIIIB		9 VIIIB		10 VIIIB		11 IB		12 IIB		13 IIIA		14 IVA		15 VA
25 Mn			26 Fe		27 Co		28 Ni		29 Cu		30 Zn		31 Ga		32 Ge		33 As
Manganese 54.938044			Iron 55.845		Cobalt 58.933194		Nickel 58.6934		Copper 63.546		Zinc 65.38		Gallium 69.723		Germanium 72.64		Antimony 121.757
43 Tc			44 Ru		45 Rh		46 Pd		47 Ag		48 Cd		49 In		50 Sn		51 Sb
Technetium (98)			Ruthenium 101.07		Rhodium 102.90550		Palladium 106.42		Silver 107.8682		Cadmium 112.414		Indium 114.818		Tin 118.710		Bismuth 208.9804
55 Re			56 Os		57 Ir		58 Pt		59 Au		60 Hg		61 Tl		62 Pb		63 Bi
Rhenium 186.207			Osmium 192.22		Iridium 192.217		Platinum 195.084		Gold 196.966569		Mercury 200.592		Thallium 204.38		Lead 207.2		Po (209)
107 Bh			108 Hs		109 Mt		110 Ds		111 Rg		112 Cn		113 Nh		114 Fl		115 Mc
Bohrium (270)			Hassium (269)		Meitnerium (278)		Darmstadtium (281)		Roentgenium (282)		Copernicium (285)		Nihonium (286)		Flerovium (289)		Moscovium (290)
61 Pm			62 Sm		63 Eu		64 Gd		65 Tb		66 Dy		67 Ho		68 Er		69 Tm
Promethium (145)			Samarium 150.36		Europium 151.964		Gadolinium 157.25		Terbium 158.92535		Dysprosium 162.500		Holmium 164.93032		Erbium 167.259		Thulium 168.93421
93 Np			94 Pu		95 Am		96 Cm		97 Bk		98 Cf		99 Es		100 Fm		101 Md
Neptunium (237)			Plutonium (244)		Americium (243)		Curium (247)		Berkelium (247)		Californium (251)		Einsteinium (252)		Fermium (257)		Mendelevium (258)

- ▶ What is the molecular weight?
- ▶ What are the numbers of
 - ▶ Protons
 - ▶ Neutron
 - ▶ Electrons

Periodic Table of the Elements

Periodic table

Number →	1	← Symbol
Name →	Hydrogen	← Atomic Weight
	1.008	

6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB
24 Cr Chromium 51.9961	25 Mn Manganese 54.938044	26 Fe Iron 55.845	27 Co Cobalt 58.933194	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.38
42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411
74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.084	79 Au Gold 196.966569	80 Hg Mercury 200.59
106 Sg Seaborgium (266)	107 Bh Bohrium (270)	108 Hs Hassium (269)	109 Mt Meitnerium (276)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)
60 Pr Praseodymium 140.90766	61 Nd Neodymium 144.242	62 Pm Promethium (145)	63 Sm Samarium 150.36	64 Eu Europium 151.964	65 Gd Gadolinium 157.25	66 Tb Terbium 158.92535
92 U Uranium 238.02891	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)

- ▶ Review the periodic table.
- ▶ Where are the periods?
- ▶ Where are the groups?

Measuring weight



Follow the instructions for weighing the sand samples.

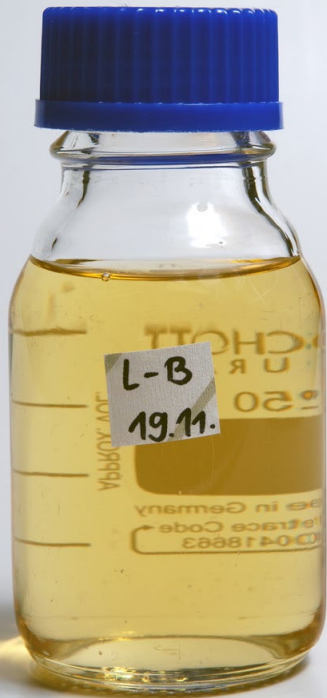


How much did your sample weigh?

Acids and bases

- ▶ Review the pH scale.
- ▶ What is pH?
- ▶ Identify where acids and bases fall on the scale.
- ▶ Try your hand at categorizing the items.





Making media

- ▶ Follow the protocol to make your own media.

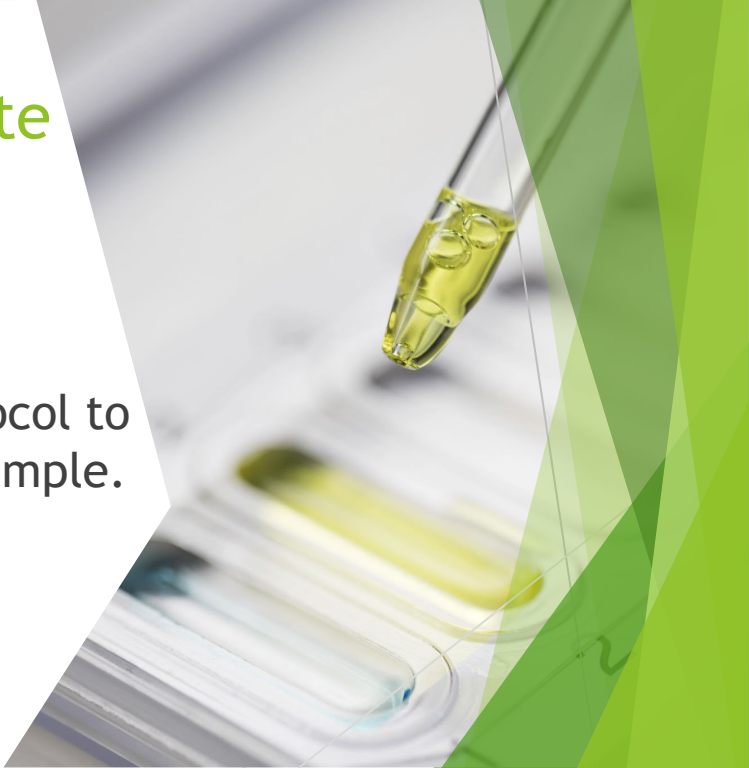
Using a spectrophotometer

- ▶ Follow the protocol to measure your sample.
- ▶ Be sure to write down your absorbance reading



Using a pipette

- ▶ Follow the protocol to measure your sample.





Using a Bacti-incinerator

- ▶ Try it out!
- ▶ Follow the protocol card.

Metric System- Conversions

► Review the cards.

► Complete them to convert using the metric system

Metric prefixes						
Prefix	Symbol	1000 ^m	10 ⁿ	Decimal	Short scale	Long scale
yotta	Y	1000 ⁸	10 ²⁴	1000000000000000000000000	septillion	quadrillion
zetta	Z	1000 ⁷	10 ²¹	100000000000000000000000	sextillion	trillion
exa	E	1000 ⁶	10 ¹⁸	10000000000000000000000	quintillion	trillion
peta	P	1000 ⁵	10 ¹⁵	1000000000000000000000	quadrillion	billiard
tera	T	1000 ⁴	10 ¹²	100000000000000000000	trillion	billion
giga	G	1000 ³	10 ⁹	10000000000000000000	billion	milliard
mega	M	1000 ²	10 ⁶	1000000	million	
kilo	k	1000 ¹	10 ³	1000	thousand	
hecto	h	1000 ^{2/3}	10 ²	100	hundred	
deca	da	1000 ^{1/3}	10 ¹	10	ten	
		1000 ⁰	10 ⁰	1	one	
deci	d	1000 ^{-1/3}	10 ⁻¹	0.1	tenth	
centi	c	1000 ^{-2/3}	10 ⁻²	0.01	hundredth	
milli	m	1000 ⁻¹	10 ⁻³	0.001	thousandth	
micro	μ	1000 ⁻²	10 ⁻⁶	0.000001	millionth	
nano	n	1000 ⁻³	10 ⁻⁹	0.000000001	billionth	milliardth
pico	p	1000 ⁻⁴	10 ⁻¹²	0.000000000001	trillionth	billionth
femto	f	1000 ⁻⁵	10 ⁻¹⁵	0.000000000000001	quadrillionth	billiardth
atto	a	1000 ⁻⁶	10 ⁻¹⁸	0.000000000000000001	quintillionth	trillionth
zepto	z	1000 ⁻⁷	10 ⁻²¹	0.000000000000000000001	sextillionth	trilliardth
yocto	y	1000 ⁻⁸	10 ⁻²⁴	0.000000000000000000000001	septillionth	quadrillionth